

Analytical derivation of the Wald–Gauss–Bonnet dark-energy sector in scale-factor variables:

$$w_{\text{DE}}(a), \, dw_{\text{DE}}/da, \, \text{and the density integral} \\ \int_a^1 3 [1 + w_{\text{DE}}(a')]/a' \, da'$$

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Abstract

We present a fully explicit, step-by-step derivation of the three quantities required to implement the Wald–Gauss–Bonnet (WGB) topological dark-energy sector as a fluid in the CLASS Boltzmann code: the equation-of-state parameter $w_{\text{DE}}(a)$, its scale-factor derivative dw_{DE}/da , and the density super-integral $\int_a^1 3 [1 + w_{\text{DE}}(a')]/a' \, da'$ that enters the fluid continuity solution $\rho_{\text{DE}}(a) = \rho_{\text{DE},0} \exp[\dots]$. All three follow from a single closed-form denominator $D(a)$ and a single kinematic identity, $dI/da = \psi(a)/a$. We use the coefficient corrected to $-24C_n$ (equivalently $+24C_n$ once the integration limits are written in the CLASS convention), rather than the $+6C_n$ appearing in the companion late-time analysis. The final results are collected in Sec. 5 in a form ready for direct transcription into `source/background.c`.

1 Starting point and conventions

The effective WGB dark-energy equation of state in redshift, as derived from the modified Friedmann equation and the Model II normalization condition $H(z=0) = H_0$, reads

$$w_{\text{DE}}(z) = -1 - \frac{8C_n \psi(z)}{3(1 - \Omega_{m0}) - 24C_n \int_0^z \frac{\psi(z')}{1 + z'} \, dz'}, \quad (1)$$

with the dimensionless amplitude $C_n \equiv \tilde{\alpha}C/H_0^2$ and the Madau–Dickinson star-formation rate

$$\psi(z) = 0.015 \frac{(1 + z)^{2.7}}{1 + [(1 + z)/2.9]^{5.6}}. \quad (2)$$

The relation between the scale factor a (with $a_0 = 1$) and redshift is

$$1 + z = \frac{1}{a}, \quad dz = -\frac{da}{a^2}, \quad \dot{z} = -\frac{\dot{a}}{a^2} = -H(1 + z), \quad (3)$$

the last of which we will use for the equation-of-state consistency check.

The star-formation rate in scale-factor form. Substituting $1 + z = 1/a$ into (2),

$$\boxed{\psi(a) = \frac{0.015 a^{-2.7}}{1 + (2.9 a)^{-5.6}}} \quad (4)$$

where we used $(1 + z)^{2.7} = a^{-2.7}$ and $[(1 + z)/2.9]^{5.6} = [1/(2.9 a)]^{5.6} = (2.9 a)^{-5.6}$.

The integral in scale-factor form. Define the redshift integral appearing in (1) and transform it to a scale-factor integral. With (3),

$$\begin{aligned}\int_0^z \frac{\psi(z')}{1+z'} dz' &= \int_{a'=1}^{a'=a} \frac{\psi(a')}{1/a'} \left(-\frac{da'}{a'^2} \right) \\ &= \int_a^1 \frac{\psi(a')}{a'} da' \equiv I(a),\end{aligned}\tag{5}$$

where in the second line we used $1/(1+z') = a'$, cancelled one power of a' , and flipped the (reversed) limits to absorb the overall minus sign. Thus

$$I(a) \equiv \int_a^1 \frac{\psi(a')}{a'} da', \quad I(1) = 0.\tag{6}$$

Note $I(a) \geq 0$ and increases monotonically as a decreases (higher redshift), since the integrand is positive.

The master identity. By the fundamental theorem of calculus applied to (6), and noting that a is the *lower* limit of integration,

$$\frac{dI}{da} = \frac{d}{da} \int_a^1 \frac{\psi(a')}{a'} da' = -\frac{\psi(a)}{a}.\tag{7}$$

This single identity drives every derivative below. (The minus sign is because differentiating with respect to the lower limit reverses the sign of the integrand evaluated at that limit.)

2 The equation of state $w_{\text{DE}}(a)$

Substituting (5) into (1) gives the equation of state as a function of the scale factor,

$$w_{\text{DE}}(a) = -1 - \frac{8C_n \psi(a)}{D(a)},\tag{8}$$

where we have introduced the *denominator function*

$$D(a) \equiv 3(1 - \Omega_{m0}) - 24C_n I(a) = 3(1 - \Omega_{m0}) - 24C_n \int_a^1 \frac{\psi(a')}{a'} da'.\tag{9}$$

Two remarks fix the CLASS sign conventions unambiguously:

- The physical normalization $D(1) = 3(1 - \Omega_{m0})$ follows immediately from $I(1) = 0$; this is the present-day value D_0 .
- In the code the spline stores $I_z(a) \equiv -I(a)$ (i.e. the negative of (6)); with that variable $D = 3(1 - \Omega_{m0}) + 24C_n I_z(a)$. The physics is identical; only the stored sign of the tabulated integral differs. We write all analytic results in terms of $I(a)$ of (6) and give the code-ready translation in Sec. 5.

Including radiation for completeness, the constant term generalizes to $3(1 - \Omega_{m0} - \Omega_{r0})$, which is what CLASS uses; we keep $3(1 - \Omega_{m0})$ in the analytic expressions and restore Ω_{r0} in the final code block.

Consistency check (optional). That (8) is the correct equation of state can be verified directly from the definition $w_{\text{DE}} = -1 - \dot{\rho}_{\text{DE}}/(3H\rho_{\text{DE}})$. With $\frac{8\pi G}{3}\rho_{\text{DE}}(z) = H_0^2(1 - \Omega_{m0}) - 8\tilde{\alpha}C I$ and $\dot{z} = -H(1+z)$ one finds $\frac{8\pi G}{3}\dot{\rho}_{\text{DE}} = 8\tilde{\alpha}CH\psi$, and dividing by $3H\rho_{\text{DE}}$ reproduces (8) with the denominator (9). This is the step at which the corrected coefficient -24 (rather than $+6$) is fixed.

3 The derivative dw_{DE}/da

We need dw_{DE}/da for the CLASS background pressure perturbation. From (8),

$$\frac{dw_{\text{DE}}}{da} = -8C_n \frac{d}{da} \left[\frac{\psi(a)}{D(a)} \right]. \quad (10)$$

Apply the quotient rule to the bracket:

$$\frac{d}{da} \left[\frac{\psi}{D} \right] = \frac{\psi'(a) D(a) - \psi(a) D'(a)}{D(a)^2}, \quad (11)$$

where $\psi'(a) \equiv d\psi/da$ and $D'(a) \equiv dD/da$.

Derivative of the denominator. Differentiating (9) and using the master identity (7),

$$D'(a) = -24C_n \frac{dI}{da} = -24C_n \left(-\frac{\psi(a)}{a} \right) = +\frac{24C_n \psi(a)}{a}. \quad (12)$$

Derivative of the star-formation rate. Write $\psi(a) = 0.015 a^{-2.7} g(a)^{-1}$ with $g(a) \equiv 1 + (2.9a)^{-5.6}$. Logarithmic differentiation gives $\psi'/\psi = -2.7/a - g'/g$, and since $g'(a) = -5.6(2.9a)^{-5.6}/a$ we have $-g'/g = \frac{5.6(2.9a)^{-5.6}}{a[1 + (2.9a)^{-5.6}]}$. Hence

$$\boxed{\psi'(a) = \psi(a) \left[-\frac{2.7}{a} + \frac{5.6(2.9a)^{-5.6}}{a(1 + (2.9a)^{-5.6})} \right]}. \quad (13)$$

This matches the closed form used in the code; it is needed only as an input to (11) and can equally be evaluated by automatic differentiation.

Assembling the derivative. Insert (12) into (11):

$$\frac{d}{da} \left[\frac{\psi}{D} \right] = \frac{\psi' D - \psi \cdot \frac{24C_n \psi}{a}}{D^2} = \frac{\psi' D - \frac{24C_n \psi^2}{a}}{D^2}. \quad (14)$$

Multiplying by $-8C_n$ as in (10),

$$\boxed{\frac{dw_{\text{DE}}}{da} = \frac{8C_n}{D(a)^2} \left[-\psi'(a) D(a) + \frac{24C_n \psi(a)^2}{a} \right]}. \quad (15)$$

Comparison with the +6 convention (why the sign flips). It is instructive to see explicitly why this is *not* obtained from the old expression by the naive replacement $6 \rightarrow 24$. In the old convention $D_{\text{old}} = 3(1 - \Omega_{m0}) + 6C_n I$, so $D'_{\text{old}} = -6C_n \psi/a$ carries the *opposite* sign to (12). Propagating that through the quotient rule flips the sign of the second bracket term. Concretely,

$$\text{old: } \frac{dw_{\text{DE}}}{da} = \frac{8C_n}{D_{\text{old}}^2} \left[-\psi' D_{\text{old}} - \frac{6C_n \psi^2}{a} \right], \quad (16)$$

whereas the corrected form (15) has $+24C_n \psi^2/a$. Both the coefficient ($6 \rightarrow 24$) and the internal sign ($- \rightarrow +$) change.

4 The density super-integral and the $\log(D/D_0)$ form

The CLASS fluid continuity equation $\rho'_{\text{DE}} = -3\mathcal{H}(1 + w_{\text{DE}})\rho_{\text{DE}}$ integrates to

$$\rho_{\text{DE}}(a) = \rho_{\text{DE},0} \exp[\mathcal{J}(a)], \quad \mathcal{J}(a) \equiv \int_a^1 \frac{3[1 + w_{\text{DE}}(a')]}{a'} da'. \quad (17)$$

We now evaluate $\mathcal{J}(a)$ in closed form.

Step 1: the integrand. From (8),

$$1 + w_{\text{DE}}(a') = -\frac{8C_n \psi(a')}{D(a')}, \quad (18)$$

so that

$$\mathcal{J}(a) = \int_a^1 \frac{3}{a'} \left(-\frac{8C_n \psi(a')}{D(a')} \right) da' = -24C_n \int_a^1 \frac{\psi(a')}{a' D(a')} da'. \quad (19)$$

Step 2: recognizing an exact logarithmic derivative. This is the crucial step. Consider $d/da' \ln D(a')$. Using (12) evaluated at a' ,

$$\frac{d}{da'} \ln D(a') = \frac{D'(a')}{D(a')} = \frac{1}{D(a')} \cdot \frac{24C_n \psi(a')}{a'} = \frac{24C_n \psi(a')}{a' D(a')}. \quad (20)$$

The integrand of (19) is therefore *exactly* the negative of a total derivative:

$$-24C_n \frac{\psi(a')}{a' D(a')} = -\frac{d}{da'} \ln D(a'). \quad (21)$$

Step 3: integrating the total derivative. Substituting (21) into (19) and applying the fundamental theorem of calculus,

$$\begin{aligned} \mathcal{J}(a) &= \int_a^1 \left(-\frac{d}{da'} \ln D(a') \right) da' = -\left[\ln D(a') \right]_{a'=a}^{a'=1} \\ &= -(\ln D(1) - \ln D(a)) = \ln D(a) - \ln D(1). \end{aligned} \quad (22)$$

With $D(1) = D_0 = 3(1 - \Omega_{m0})$,

$$\boxed{\mathcal{J}(a) = \int_a^1 \frac{3[1 + w_{\text{DE}}(a')]}{a'} da' = \ln \frac{D(a)}{D_0} = +1 \cdot \ln \frac{D(a)}{D_0}.} \quad (23)$$

Why the prefactor is +1, not -4. The general structure is transparent from Step 2. Write the denominator with a generic coefficient, $D_\beta = 3(1 - \Omega_{m0}) + \beta C_n I$, and keep the numerator coefficient of $1 + w$ fixed at the value κ dictated by the model (here the same κ that appears as $-\kappa C_n \psi/D$). Then

$$\frac{d}{da'} \ln D_\beta = \frac{D'_\beta}{D_\beta} = \frac{-\beta C_n dI/da'}{D_\beta} = \frac{\beta C_n \psi(a')/a'}{D_\beta}, \quad (24)$$

and the super-integral becomes

$$\mathcal{J} = -\kappa C_n \int_a^1 \frac{\psi}{a' D_\beta} da' = -\frac{\kappa}{\beta} \left(\ln D_\beta(a) - \ln D_{\beta,0} \right) = +\frac{\kappa}{\beta} \ln \frac{D_\beta(a)}{D_{\beta,0}}. \quad (25)$$

Thus the prefactor is the ratio κ/β of the numerator coefficient to the denominator coefficient:

Convention	κ (numerator of $3(1+w)$)	β (denominator)	prefactor κ/β
old ($+6C_n$)	24	6	-4
corrected ($-24C_n$)	24	-24	+1

In the old convention $\kappa/\beta = 24/6 = 4$, and the two D -signs conspire to give the code's $-4\ln(D/D_0)$. In the corrected convention $\kappa/\beta = 24/(-24) = -1$, so $\mathcal{J} = +\ln(D/D_0)$. The prefactor therefore changes from -4 to $+1$: not a rescaling but a sign-and-magnitude change, exactly as for dw/da in Sec. 3.

Sanity limit. As $C_n \rightarrow 0$, $D(a) \rightarrow D_0$ and $\mathcal{J}(a) \rightarrow \ln 1 = 0$, hence $\rho_{\text{DE}}(a) \rightarrow \rho_{\text{DE},0} = \text{const}$: the cosmological-constant (Λ CDM) limit is recovered, as required.

5 Summary and code-ready expressions

All three quantities are built from the single denominator function

$$D(a) = 3(1 - \Omega_{m0} - \Omega_{r0}) - 24 C_n I(a), \quad I(a) = \int_a^1 \frac{\psi(a')}{a'} da', \quad D_0 = 3(1 - \Omega_{m0} - \Omega_{r0}), \quad (26)$$

and the master identity $dI/da = -\psi(a)/a$. The results are

$$w_{\text{DE}}(a) = -1 - \frac{8C_n \psi(a)}{D(a)}, \quad (27)$$

$$\frac{dw_{\text{DE}}}{da} = \frac{8C_n}{D(a)^2} \left[-\psi'(a) D(a) + \frac{24C_n \psi(a)^2}{a} \right], \quad (28)$$

$$\mathcal{J}(a) = \int_a^1 \frac{3[1 + w_{\text{DE}}(a')]}{a'} da' = \ln \frac{D(a)}{D_0}, \quad (29)$$

with $\psi(a)$ from (4) and $\psi'(a)$ from (13).

Translation to the `class_wgb` variables. The code stores $I_z(a) \equiv -I(a)$ (so that the tabulated array is the negative of (6)). In terms of the code variable $\text{Iz} = I_z(a)$,

$$D(a) = 3(1 - \Omega_{m0} - \Omega_{r0}) + 24 C_n \text{Iz}, \quad (30)$$

and the three implementations become, with Da , Da0 , $\text{psi_a} = \psi(a)$, $\text{dpsi_da} = \psi'(a)$:

$$\begin{aligned} \text{w_fld} &= -1.0 - (8 \text{Cn} \cdot \text{psi_a})/\text{Da}; \\ \text{dw_over_da_fld} &= (8 \text{Cn}/(\text{Da} \cdot \text{Da})) \times [-\text{dpsi_da} \cdot \text{Da} + 24 \text{Cn} \cdot \text{psi_a} \cdot \text{psi_a}/a]; \\ \text{integral_fld} &= +1.0 \cdot \log(\text{Da}/\text{Da0}). \end{aligned}$$

Relative to the previously released code these amount to: (i) the denominator coefficient $6 \rightarrow 24$ with the code's “`-denom2`” becoming “`+ 24 Cn Iz`”; (ii) in `dw_over_da_fld`, both the coefficient $6 \rightarrow 24$ and the sign of the ψ^2 term flipping from $-$ to $+$; and (iii) the density prefactor $-4 \rightarrow +1$. The generic fluid solution $\rho_{\text{DE}} = \rho_{\text{DE},0} \exp(\text{integral_fld})$ and the positivity test $D > 0$ are unchanged in form.